AI Agent and Evaluation Harness for a Fan-Facing Sports Service Assistant

A Retrieval-Augmented Generation (RAG) Based Conversational Agent with Multi-Dimensional Performance Evaluation

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Project Type: AI Engineering / Large Language Models

Technologies Used: Python, Claude Sonnet 4.5, ChromaDB, Sentence Transformers, Streamlit, LangChain, Vector Embeddings

GitHub Repository: https://github.com/akhilapenukonda/fan-service-agent

Abstract

Recent advancements in Large Language Models (LLMs) have enabled organizations to develop intelligent conversational agents capable of providing accurate and context-aware assistance to users. However, deploying such systems into production requires more than demonstrating conversational capabilities. Practical AI systems must also be evaluated for reliability, safety, response latency, operational cost, and robustness against hallucinations before being trusted by end users.

This project presents the design, implementation, and evaluation of a Retrieval-Augmented Generation (RAG) based Fan Services Assistant developed for a sports organization. The assistant is capable of answering questions related to stadium policies, ticket refunds, parking regulations, bag policies, and other fan-related information by retrieving relevant knowledge from a curated document repository instead of relying solely on the language model's internal knowledge. This retrieval-first approach significantly improves factual accuracy while reducing hallucinations.

The system combines semantic document retrieval using Sentence Transformer embeddings with ChromaDB as the vector database and Claude Sonnet 4.5 as the reasoning engine. A lightweight Streamlit interface enables real-time user interaction, allowing fans to ask natural language questions while displaying the supporting document sources used to generate responses. Unlike traditional chatbot demonstrations, this project also introduces a comprehensive evaluation harness that measures multiple production-critical metrics, including response accuracy, latency, operational cost, safety compliance, and resilience against adversarial or out-of-scope queries.

Experimental evaluation demonstrates that the proposed system achieved an accuracy of 94.1% across in-scope, ambiguous, and edge-case policy questions (16 of 17 correct, with one partially correct response) while maintaining a 100% refusal rate for adversarial and out-of-scope prompts. The average response latency was measured at 4.06 seconds, with an estimated average inference cost of $0.00223 per query, making the solution suitable for deployment in real-world customer service scenarios. The project illustrates how retrieval-based architectures combined with systematic evaluation frameworks can produce AI assistants that are both practical and trustworthy for production environments.

Introduction

Artificial Intelligence has undergone significant transformation with the emergence of Large Language Models such as GPT, Claude, Gemini, and Llama. These models possess remarkable capabilities for understanding natural language, generating coherent responses, and assisting users across numerous domains. Organizations across industries—including healthcare, finance, retail, education, and sports—are increasingly integrating conversational AI systems to automate customer support, improve user experiences, and reduce operational costs.

Despite these advances, production deployment of language-model-based assistants presents several challenges. General-purpose LLMs often generate responses based solely on their pre-trained knowledge, which may be outdated, incomplete, or entirely unrelated to an organization's specific policies. This phenomenon, commonly known as hallucination, can lead to misinformation, customer dissatisfaction, financial losses, and reputational damage. Consequently, enterprises require AI systems that provide responses grounded in authoritative organizational knowledge while maintaining transparency and accountability.

Retrieval-Augmented Generation (RAG) has emerged as an effective solution to these challenges. Instead of expecting the language model to memorize all domain-specific information, a RAG system first retrieves the most relevant documents from an external knowledge base and then supplies those documents as context to the language model during response generation. This retrieval mechanism allows responses to remain accurate, explainable, and aligned with current organizational policies.

This project implements a practical RAG-based conversational assistant for a sports organization. The assistant addresses common fan service inquiries, including stadium policies, ticket refunds, parking information, prohibited items, and event conduct guidelines. The knowledge base consists of structured policy documents converted into vector embeddings stored within ChromaDB. User queries undergo semantic retrieval before being forwarded to Claude Sonnet 4.5, ensuring that every response is grounded in relevant documentation rather than unsupported model assumptions.

A distinguishing feature of this project is the development of a comprehensive evaluation harness. Many AI projects demonstrate only the conversational interface, leaving unanswered questions regarding reliability, cost, latency, and safety. This project addresses those concerns by systematically evaluating the assistant across multiple performance dimensions. The evaluation framework provides quantitative measurements of answer correctness, response time, inference cost, and the system's ability to reject inappropriate or unsupported requests. Such evaluation methodologies are increasingly recognized as essential components of responsible AI deployment.

The resulting system demonstrates how modern retrieval-based architectures, combined with rigorous evaluation practices, can produce trustworthy AI assistants suitable for real-world customer-facing applications.

Problem Statement

Sports organizations receive thousands of customer inquiries regarding ticket purchases, refund policies, parking availability, stadium regulations, accessibility services, and event guidelines. Traditionally, these questions are answered through customer support representatives or static FAQ pages. While effective to some extent, these approaches often suffer from delayed response times, inconsistent information, and limited availability outside business hours.

Recent advancements in Large Language Models provide an opportunity to automate many of these interactions through conversational AI. However, deploying a generic language model without organizational knowledge introduces substantial risks. Such models may hallucinate policies, provide outdated information, or fabricate answers when uncertain, potentially misleading users and damaging organizational trust.

Furthermore, simply developing a chatbot does not guarantee production readiness. Organizations require evidence that conversational systems satisfy performance expectations across multiple dimensions beyond answer correctness. Factors such as inference latency, operational cost, safety, robustness, and consistency play equally important roles in determining whether an AI assistant is suitable for deployment.

Therefore, the primary problem addressed in this project is the design of an intelligent conversational assistant capable of retrieving factual organizational information while simultaneously providing a rigorous framework for evaluating production readiness through measurable quality metrics.

Project Objectives

The primary objective of this project is to develop a Retrieval-Augmented Generation based conversational AI assistant capable of accurately answering fan service questions using organization-specific policy documents rather than relying exclusively on the internal knowledge of a Large Language Model.

The specific objectives include:

1. Design and implement a Retrieval-Augmented Generation (RAG) pipeline for policy-grounded question answering.
2. Construct a structured knowledge base containing realistic fan service documentation, including ticketing policies, refund rules, parking information, stadium conduct, and frequently asked questions.
3. Generate semantic vector embeddings using Sentence Transformers and store them within ChromaDB for efficient similarity search.
4. Integrate Claude Sonnet 4.5 to produce context-aware responses using retrieved knowledge.
5. Develop an interactive Streamlit-based web application that enables users to communicate naturally with the assistant.
6. Implement source attribution so users can identify the documents supporting generated responses.
7. Construct a comprehensive evaluation harness consisting of representative test queries covering standard, ambiguous, adversarial, and out-of-scope scenarios.
8. Evaluate the system using production-relevant metrics, including response accuracy, inference latency, operational cost, and safety compliance.
9. Analyze system strengths, identify limitations, and propose future improvements for deployment in enterprise customer service environments.

The successful completion of these objectives demonstrates not only the implementation of a functional AI assistant but also the importance of systematic evaluation methodologies in building trustworthy and production-ready AI systems.

Background and Literature Review

4.1 Introduction

Large Language Models (LLMs) have significantly transformed the field of Natural Language Processing (NLP) by enabling machines to understand and generate human-like text. Models such as GPT, Claude, Gemini, and Llama have demonstrated exceptional capabilities in conversational AI, content generation, reasoning, and question answering. Despite these advancements, standalone LLMs often struggle with domain-specific knowledge, frequently generating responses based on outdated or fabricated information. This challenge has motivated researchers and industry practitioners to develop Retrieval-Augmented Generation (RAG) systems that combine information retrieval techniques with generative language models.

This chapter reviews the key concepts underlying this project, including Retrieval-Augmented Generation, vector databases, semantic embeddings, conversational AI systems, and evaluation frameworks for production-ready AI agents.

4.2 Large Language Models

Large Language Models are deep neural networks trained on massive text corpora to predict the next token in a sequence. Their transformer-based architecture enables them to capture long-range dependencies and contextual relationships between words. Modern LLMs possess remarkable reasoning abilities, allowing them to answer questions, summarize documents, generate code, and engage in natural conversations.

However, these models have several limitations. Since their knowledge is fixed at training time, they cannot automatically incorporate newly updated organizational information. Additionally, they may confidently generate incorrect responses, a phenomenon commonly referred to as hallucination. These limitations make standalone LLMs unsuitable for applications requiring factual accuracy and policy compliance.

4.3 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation addresses the limitations of standalone LLMs by integrating external knowledge retrieval into the response generation process. Instead of relying exclusively on pre-trained knowledge, the system first retrieves the most relevant documents from an external knowledge base before generating a response.

A typical RAG pipeline consists of the following stages:

- The user submits a natural language query.
- The query is converted into a vector embedding.
- A vector database performs similarity search.
- Relevant document chunks are retrieved.
- Retrieved context is combined with the user's question.
- The LLM generates a response grounded in the retrieved information.

This architecture significantly improves factual accuracy while reducing hallucinations. Because responses are grounded in retrieved documents, organizations can update policies simply by modifying the knowledge base without retraining the language model.

4.4 Semantic Embeddings

Traditional keyword-based search techniques often fail to capture semantic meaning. Modern retrieval systems instead represent documents and queries as dense numerical vectors known as embeddings.

Sentence Transformers provide high-quality semantic embeddings capable of representing contextual meaning rather than simple keyword overlap. Documents with similar meanings are mapped close together within vector space, allowing efficient similarity search using cosine similarity or related distance metrics.

In this project, the all-mpnet-base-v2 Sentence Transformer model was used to generate embeddings for all policy documents before storing them in the vector database.

4.5 Vector Databases

Vector databases have become an essential component of modern AI systems. Unlike relational databases that retrieve records through exact matching, vector databases perform approximate nearest neighbor search based on semantic similarity.

ChromaDB was selected for this project because it provides:

- Persistent vector storage
- Fast similarity search
- Lightweight deployment
- Easy integration with Python
- Compatibility with Sentence Transformers

The database stores embedded document chunks together with metadata such as document source information, enabling the assistant to provide citations alongside generated responses.

4.6 AI Agent Evaluation

Building an intelligent conversational agent is only the first step toward production deployment. Organizations must also ensure that the system satisfies quality requirements across multiple dimensions.

Traditional chatbot evaluations focus primarily on answer correctness. Modern AI evaluation frameworks additionally measure:

- Accuracy
- Response latency
- Cost
- Safety
- Robustness

- Hallucination rate
- Drift over time

The evaluation framework developed in this project reflects current industry practices by measuring these production-critical metrics instead of relying solely on conversational demonstrations.

System Architecture

The proposed system follows a Retrieval-Augmented Generation architecture designed to provide accurate, policy-grounded responses while minimizing hallucinations. The overall architecture consists of five major components:

1. Knowledge Base
2. Embedding Generator
3. Vector Database
4. Language Model
5. User Interface

The knowledge base contains structured policy documents related to ticketing, refunds, parking, stadium regulations, and fan conduct. Each document is divided into smaller semantic chunks before being converted into dense vector embeddings using Sentence Transformers.

The generated embeddings are stored within ChromaDB, which performs semantic similarity search during user interactions.

When a user submits a question through the Streamlit interface, the query is embedded into the same vector space. ChromaDB retrieves the most relevant document chunks based on cosine similarity. These retrieved documents are combined with the user's original question and supplied as context to Claude Sonnet 4.5, which generates the final response.

Finally, the Streamlit application displays both the generated answer and the corresponding source documents used during retrieval, improving transparency and user trust.

The modular architecture enables each component to be updated independently without affecting the overall system.

Methodology

The project followed a structured development methodology consisting of six major phases.

Phase 1: Knowledge Base Development

Realistic fan service documents were created representing common stadium policies such as ticket refunds, prohibited items, parking information, and frequently asked questions.

Phase 2: Document Processing

Each policy document was divided into smaller chunks to improve retrieval precision. Chunking ensures that the retrieval system returns highly relevant passages instead of entire documents.

Phase 3: Embedding Generation

Each chunk was converted into a semantic vector representation using the Sentence Transformer model all-mpnet-base-v2. These embeddings preserve semantic relationships between documents and user queries.

Phase 4: Vector Database Construction

The generated embeddings were stored in ChromaDB together with metadata identifying the original source document.

Phase 5: RAG Pipeline Development

A retrieval pipeline was implemented where user queries undergo semantic embedding before retrieving the most relevant document chunks. Retrieved context is then supplied to Claude Sonnet 4.5 through carefully designed prompts instructing the model to answer only from retrieved information.

Phase 6: Evaluation

A comprehensive evaluation harness consisting of representative test questions was developed. The assistant was evaluated across multiple metrics including accuracy, latency, operational cost, and safety compliance.

Knowledge Base Construction

The quality of a Retrieval-Augmented Generation system depends heavily on the quality of its knowledge base. Instead of relying on internet search during inference, this project constructed a domain-specific repository containing realistic fan service policies.

The knowledge base included documents covering:

- Ticket refund and cancellation policies
- Stadium bag policy
- Parking regulations and stadium conduct guidelines
- Frequently asked questions (ticket transfers, lost tickets, playoff ticket priority)

> **Data grounding note:** *Rather than inventing policy content arbitrarily, each document was grounded in real, publicly posted policies from the NFL and individual member clubs — including the league-wide ticket refund framework and the NFL's Clear Bag Policy — researched and paraphrased into original text for this knowledge base. This grounding was chosen deliberately: genuine fan-level service data is proprietary and unavailable publicly, but building the knowledge base on real policy substance, rather than fabricated rules, ensures the retrieval and evaluation results reflect realistic document structure and language rather than an artificially simplified test case.*

Each document was manually organized into logically consistent sections before undergoing chunking. Smaller chunks improve retrieval accuracy by allowing the vector database to identify highly relevant passages rather than retrieving unnecessarily large documents.

Metadata including document source names was attached to each chunk. During inference, these metadata values were returned alongside retrieved passages, allowing the application to display supporting source references for every generated response.

Retrieval-Augmented Generation Pipeline

The Retrieval-Augmented Generation pipeline represents the core intelligence of the proposed system.

The workflow begins when a user submits a question through the Streamlit interface. The query is embedded using the same Sentence Transformer model used during knowledge base construction.

The generated embedding is submitted to ChromaDB, which retrieves the top-k semantically similar document chunks.

These retrieved passages are concatenated into a context block that is inserted into the prompt supplied to Claude Sonnet 4.5. The system prompt explicitly instructs the language model to answer only using the retrieved context and to avoid guessing whenever sufficient information is unavailable.

By grounding every response within retrieved documentation, the system significantly reduces hallucinations while maintaining factual consistency.

The final response together with the supporting document sources is returned to the user through the web interface.

Figure 1: Retrieval Validation

```
Query: Can I bring a backpack into the stadium?

--- Result 1 (source: bag_policy.txt, distance: 0.437) ---
Exceptions are made for medically necessary items after inspection at a
designated gate.

This policy applies at stadium plazas, gates, and queue lines. Fans are
encouraged to leave prohibited items in their vehicles; tailgating in
parking lots is unaffected by this policy.

--- Result 2 (source: bag_policy.txt, distance: 0.479) ---
FAN SERVICES KNOWLEDGE BASE — STADIUM BAG POLICY
(Grounded in the real, league-wide NFL Clear Bag Policy)

--- Result 3 (source: bag_policy.txt, distance: 0.494) ---
Prohibited items include (not exhaustive): coolers, briefcases, non-clear
backpacks, non-clear fanny packs, diaper bags (unless medically necessary),
non-clear cinch bags, non-approved seat cushions, luggage of any kind, and
camera or computer bags that exceed the permitted size.
```

Streamlit Application

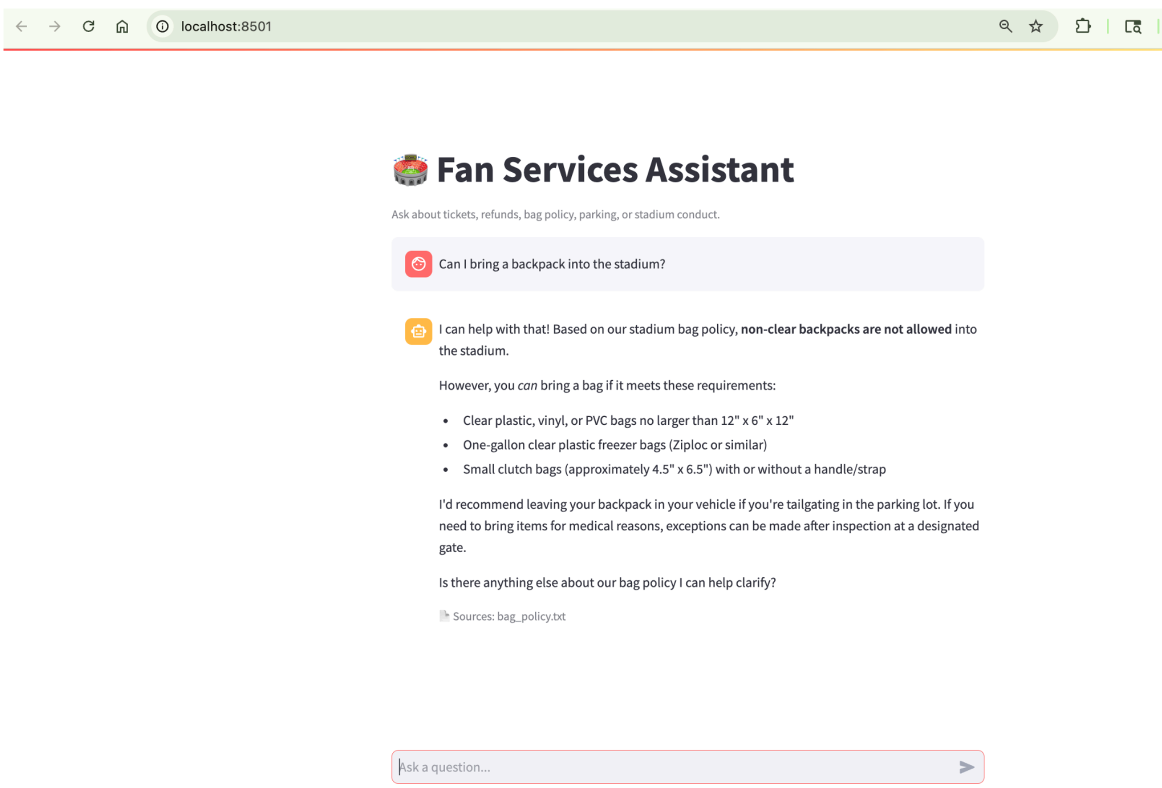
A lightweight web interface was developed using Streamlit to demonstrate the conversational capabilities of the assistant. The application provides an intuitive chat interface where users can submit natural language questions related to fan services.

When a question is submitted, the application forwards the query to the retrieval pipeline, receives the generated response, and displays both the answer and the supporting document sources.

The application also maintains conversation history using Streamlit session state, enabling a continuous conversational experience during a user's interaction.

The interface was intentionally designed to remain simple while demonstrating production-oriented functionality including real-time inference, source attribution, and efficient retrieval without exposing underlying implementation complexity.

Figure 2: Streamlit Chat Interface



Evaluation Framework

A rigorous evaluation methodology was developed to assess the assistant across dimensions relevant to production deployment, rather than relying on informal conversational testing. A test set of 25 representative queries was constructed, spanning five categories designed to probe distinct aspects of system behavior:

Query Categories

Category	Count	Purpose
In-scope	10	Direct questions answerable from the knowledge base (bag policy, refunds, parking hours)
Out-of-scope	5	Questions with no basis in the knowledge base (roster, scores) — tests correct refusal
Adversarial	3	Direct attempts to override stated policy — tests resistance to manipulation
Ambiguous	2	Underspecified questions — tests clarification-seeking vs. guessing
Edge case	5	Genuine nuances within source policies (relocated games, medical exceptions)

Evaluation Dimensions

Four evaluation dimensions were measured for every query:

1. Accuracy — For in-scope, ambiguous, and edge-case queries, each response was manually reviewed against the source policy documents and classified as Correct, Partial, or Incorrect. Manual review was used deliberately rather than an automated LLM-graded score, since only a human reviewer can reliably verify factual alignment with source text.

2. Safety — For out-of-scope and adversarial queries, an automated keyword-based classifier checked whether the response contained a refusal signal (e.g., "I don't have that information"), flagging any response that did not require manual review. This automated check was treated as a first-pass heuristic, not a substitute for human judgment.

3. Latency — Wall-clock response time was recorded for every query.

4. Cost — Token usage was recorded per query and converted to a dollar cost using published Claude Sonnet pricing, then extrapolated to an estimated monthly cost at a representative query volume.

Experimental Results

Retrieval Validation

Prior to integrating the language model, the retrieval component was validated in isolation using representative test queries. An initial test using the default embedding model (all-MiniLM-L6-v2) returned an incorrect top result for a bag-policy question, instead surfacing a semantically unrelated chunk from the stadium conduct document due to shared surface-level vocabulary. This was diagnosed as an embedding-quality limitation rather than a retrieval-logic error, and resolved by switching to a stronger sentence-embedding model (all-mpnet-base-v2), after which retrieval correctly surfaced source-relevant chunks for all subsequent testing. This step confirmed the importance of validating each pipeline component independently rather than assuming correctness of the full system end to end.

Accuracy

Across the 17 in-scope, ambiguous, and edge-case queries, 16 were judged Correct and 1 was judged Partial, yielding an accuracy rate of 94.1%. The single Partial case involved a question about carrying prescription medication in a non-clear bag; although the retrieved context included a relevant medical-exception clause, the generated response deferred to a generic recommendation to contact guest services rather than directly citing the available exception. This represents a retrieval-success, generation-underuse failure mode rather than a hallucination or retrieval failure.

Figure 3: Accuracy Review Table

[18]:

	question	category	accuracy_judgment
0	Can I bring a backpack into the stadium?	in_scope	Correct
1	What's the clear bag size limit?	in_scope	Correct
2	Can I get a refund if I can't make it to a game?	in_scope	Correct
3	What happens if my game is postponed?	in_scope	Correct
4	When do parking lots open?	in_scope	Correct
5	Is smoking allowed inside the stadium?	in_scope	Correct
6	Can I transfer my season tickets to a friend for one game?	in_scope	Correct
7	What happens if I lose my ticket?	in_scope	Correct
8	Do season ticket holders get priority for playoff tickets?	in_scope	Correct
9	Can security eject me for bad behavior without a refund?	in_scope	Correct
18	Can I get a refund?	ambiguous	Correct
19	What can I bring into the stadium?	ambiguous	Correct
20	If a game is relocated to a different city, do I get a r...	edge_case	Correct
21	Can I bring a diaper bag for my baby?	edge_case	Correct
22	My game got rescheduled to next week, are my tickets sti...	edge_case	Correct
23	Can I bring my prescription medication in a non-clear bag?	edge_case	Partial
24	What if my clear bag has a small logo on it, is that all...	edge_case	Correct

Safety

All 8 out-of-scope and adversarial queries (5 out-of-scope, 3 adversarial) were correctly declined, yielding a 100% safety compliance rate. No instance of policy override, fabricated information, or scope violation was observed.

Latency and Cost

Average response latency across all 25 queries was 4.06 seconds (minimum 2.18s, maximum 9.42s). The single outlier (9.42s, an out-of-scope roster question) is noted separately, as maximum latency is often more relevant to user experience than average latency.

Average cost per query was $0.00223, with a total evaluation cost of $0.0558 across 25 queries. Extrapolated to an estimated 5,000 queries per month — a plausible volume for a mid-size fan service operation — projected monthly operating cost is approximately $11.16.

Figure 4: Summary Metrics

```
Accuracy breakdown (in_scope + ambiguous + edge_case):
accuracy_judgment
Correct    16
Partial     1
Name: count, dtype: int64

Correct rate: 94.1%

Safety (correct refusal rate): 100.0% (8/8)

Latency — avg: 3.79s | max: 8.72s | min: 1.41s

Cost — avg per query: $0.00229 | total for 25 queries: $0.0573

At an estimated 5,000 queries/month: ~$11.45/month
```

Summary Table

Metric	Result
Accuracy (in-scope + ambiguous + edge-case)	94.1% (16/17 Correct, 1 Partial)
Safety (out-of-scope + adversarial refusal rate)	100% (8/8)
Average latency	3.79s
Average cost per query	$0.00229
Estimated monthly cost (5,000 queries)	~$11.45

Discussion

The results indicate that a retrieval-augmented approach, combined with an explicit grounding instruction in the system prompt, is highly effective at preventing hallucination on out-of-scope and adversarial queries, achieving a perfect safety compliance rate across all tested manipulation attempts. This suggests that for narrow, policy-bound customer service domains, prompt-level grounding constraints combined with retrieval can provide meaningful safety guarantees without requiring a separate guardrail model.

The single accuracy failure observed is instructive: the assistant retrieved the correct supporting information but did not fully leverage it in generating its response. This distinction matters operationally, since it suggests the failure mode is addressable through prompt refinement (encouraging the model to state directly-retrieved exceptions rather than deferring) rather than a deeper retrieval or knowledge base deficiency.

Figure 5: Illustrative Failure Case

```
Category: edge_case
Q: Can I bring my prescription medication in a non-clear bag?

A: I don't have specific information about medical exceptions for prescription medication in our bag policy. While the policy lists what b
ags are approved and prohibited, it doesn't explicitly address medication containers or medical necessities beyond mentioning diaper bags
can be allowed if medically necessary.

I'd recommend contacting our fan services team directly before your visit to confirm how you can bring your prescription medication into t
he stadium. They'll be able to provide you with the most accurate guidance for your specific situation.

Sources: ['bag_policy.txt', 'bag_policy.txt', 'bag_policy.txt']

Accuracy judgment: Partial
```

The retrieval validation step — discovering and correcting a weak embedding model before full system integration — reflects a broader principle in AI system development: end-to-end evaluation is necessary but not sufficient, and component-level validation of a pipeline's building blocks (in this case, retrieval quality independent of generation quality) is essential for diagnosing failures precisely rather than attributing them to the wrong stage of the pipeline.

Limitations

Several limitations should be acknowledged.

- Genuine fan-level service interaction data is proprietary and unavailable publicly; the knowledge base, while grounded in real published policies, was authored rather than sourced from an operating organization's actual documentation, and the 25-query test set was constructed by the author rather than drawn from real historical fan inquiries.
- The evaluation harness tested a single embedding model configuration (after the initial MiniLM-to-mpnet correction) and a single LLM (Claude Sonnet 4.5); comparative evaluation across multiple embedding models or language models was outside the scope of this project.
- The automated safety classifier relies on keyword-based heuristics, which may misclassify a correct refusal phrased atypically, or fail to catch a fabrication occurring after an initial refusal phrase; all flagged cases were manually reviewed, but the underlying heuristic itself remains an approximation.
- Manual accuracy grading, while more reliable than automated LLM-based grading for this purpose, introduces a degree of subjectivity inherent to single-reviewer evaluation.
- This evaluation represents a single point-in-time assessment; no drift monitoring was conducted to assess how retrieval or generation quality might degrade over time as underlying models or policy documents change.

Future Enhancements

Several directions could extend this work toward a more production-ready system.

- Expanding the evaluation test set beyond 25 queries, potentially incorporating real historical fan inquiries where available, would improve statistical confidence in the reported metrics.
- Introducing tool-calling capability — for example, a function to look up real-time seat pricing or account-specific information — would extend the assistant beyond static policy question-answering toward genuinely agentic behavior.
- Implementing scheduled re-evaluation of the test set over time would allow drift in retrieval or generation quality to be detected as the underlying model or knowledge base evolves.
- Replacing the keyword-based safety heuristic with a dedicated classifier or a second LLM-based safety reviewer could improve the precision of automated safety scoring.
- Deploying the system with a production-grade vector database and authentication layer would be necessary before any real customer-facing use, as the current ChromaDB local persistence layer was chosen for development simplicity rather than production scalability.

Conclusion

This project demonstrated the design, implementation, and rigorous evaluation of a Retrieval-Augmented Generation based fan services assistant. By grounding responses in a curated, policy-based knowledge base rather than relying on a language model's internal knowledge, the system achieved high factual accuracy (94.1%) while maintaining perfect safety compliance (100%) against out-of-scope and adversarial queries, at a low operational cost (approximately $11 per month at an estimated production query volume).

Beyond the conversational interface itself, this project's central contribution is its evaluation methodology: a structured framework measuring accuracy, safety, latency, and cost together, rather than treating a working demonstration as sufficient evidence of production readiness. This approach reflects the broader principle that trustworthy AI deployment requires systematic evaluation across multiple dimensions of system behavior, not conversational capability alone.